

Dielectric Materials

4.1 INTRODUCTION

Dielectrics are the insulating materials having electric dipole moment permanently or temporarily by applying the electric field. These are mainly used to store electrical energy and used as electrical insulators. All dielectrics are electrical insulators, but all electrical insulators need not to be dielectrics.

Dielectrics are non - metallic materials of high specific resistance and have negative temperature coefficient of resistance.

4.2 BASIC DEFINITIONS

4.2.1 Electric flux density (D)

The number of electric lines passing through the unit area of cross section.

$$D = [\phi / A] \quad \text{Unit: Coulomb m}^{-2}$$

4.2.2 Permittivity [ϵ]

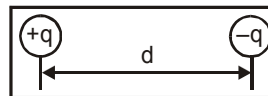
It is the ratio of electric displacement in a dielectric medium to the applied electric field strength. $\epsilon = \frac{D}{E}$ for a dielectric medium $\epsilon_r \gg 1$

4.2.3 Dipole moment (μ)

Dipole moment is defined as the product of charge and distance between two charges.

$$\mu = q \times d$$

Unit : Coulomb meter.



4.2.4 Polarization

The separation of negative and positive charges is called polarization. i.e., the i.e., process of producing electric dipoles by an electric field is called polarization

$$\mu = \alpha E$$

where α - is called Polaris ability.

4.2.5 Polarization vector

If “ μ ” is the average dipole moment per molecule and “N” is the number of molecules per unit volume then the polarization of the solid is given by the polarization vector P and it can be written as

$$P = \frac{N\mu}{V} = N\mu \quad [\text{for unit volume } V=1]$$

The polarization vector is the dipole moment per unit volume of the dielectric material.

4.2.6 Polar and Non-polar Molecules

Polar Molecules

Polar Molecules which are having permanent dipole moment even in the absence of an applied field are called polar molecules.

Example: H_2O , HCl , CO .

Non-polar Molecules

Molecules which do not have permanent dipole moment, but they have induced dipole moment in the presence of applied electric field are called non - polar molecules.

Example: O_2 , H_2 , N_2

4.2.7 Dielectric Constance (Or) Relative Permittivity (ϵ_r)

The dielectric characteristics of a material are determined by the dielectric constant (or) relative permittivity of the material

It is defined as the ratio of the permittivity of the medium to the permittivity of the free space.

$$\epsilon_r = \frac{\text{Permittivity of the medium}}{\text{Permittivity of free space}}$$

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} \quad \text{where} \quad \epsilon_0 = 8.854 \times 10^{-12} \text{ F/m.}$$

4.2.8 Electric Susceptibility (χ_e)

The polarization vector P is proportional to the total applied electric field intensity E and is in the same direction of E. Therefore, the polarization vector can be written as

$$P = \epsilon_0 \chi_e E$$

Where χ_e is electric susceptibility

Relation between Electric Susceptibility (χ) and relative permittivity (ϵ_r)

We know that $P \propto E$

$$P = \epsilon_0 \chi_e E$$

$$\chi_e = \frac{P}{\epsilon_0 E}$$

$$\text{But} \quad \epsilon_r = 1 + \chi_e$$

$$\chi_e = \epsilon_r - 1$$

$$\therefore \text{ We can write } P = \epsilon_0 E (\epsilon_r - 1)$$

It is a measure of characteristic of every dielectric.

4.2.9 Different between Polar and Non - Polar molecules

<i>S.No</i>	<i>Polar molecules</i>	<i>Non - Polar molecules</i>
1.	These molecules have permanent dipole moment even in the absence of on applied field.	These molecules do not have permanent dipole moment
2.	The polarization of polar molecules is highly temperature dependent.	The polarization in these type of molecules is independent of temperature.
3.	There is absorption or emission in the infrared range for these molecules <i>Example:</i> H ₂ O, HC1, CO	There is no absorption of emission in infrared range for these molecules <i>Example:</i> O ₂ , H ₂ , N ₂

4.3 POLARIZATION MECHANISMS INVOLVED IN A DIELECTRIC MATERIAL

There are four different types of polarization.

1. Electronic (or) induced polarization
2. Ionic (or) atomic polarization
3. Orientation (or) dipolar polarization
4. Space - Charge (or) interfacial polarization

4.3.1 Electronic Polarization

Electronic Polarization occurs due to the displacement of positively charged nucleus and negatively charged electron in opposite directions by an external electric field. It creates a dipole moment in the dielectric. This is called electronic polarization.

The induced dipole moment, $\mu_e = \alpha_e E$... (1)

where α_e - Electronic polarisability.

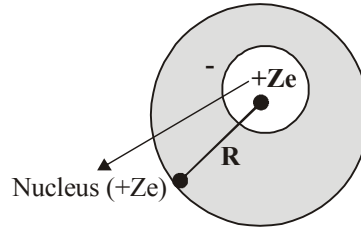
E - Electric field Strength.

It is proportional to volume of the atoms and is independent of temperature.

4.3.2 Calculation of electronic polarization α_e

Consider an atom of a dielectric material of nuclear charge (Ze). The electrons of charge ($-Ze$) are distributed uniformly throughout the atom (Sphere) of radius R as shown in figure.

With out field



(a) Without Field

Fig.4.1 Electronic polarization

$$\left. \begin{array}{l} \text{Charge density of the charged} \\ \text{sphere (electrons) of radius R} \end{array} \right\} = \frac{\text{Total charge of sphere with radius R}}{\text{Volume}}$$

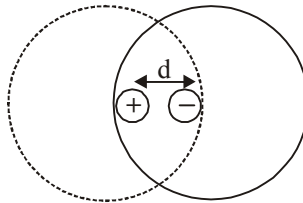
$$= \frac{-Ze}{\frac{4}{3}\pi R^3}$$

$$= \frac{-3Ze}{4\pi R^3} \quad \dots (1)$$

With field

When the atom of the dielectric is placed in an electric field (E), two types of forces are arise.

- 1) **Lorentz force:** Force which separates electrons and positive nucleus due to applied field.
- 2) **Coulomb force:** An attractive force which is produced after separation.



(b) With Field

Fig.4.2 Electronic polarization

At equilibrium

$$\text{Lorentz force} = \text{Coulomb force}$$

$$\begin{aligned}\text{Lorentz force} &= (\text{charge of the electron}) \times (\text{intensity of applied Field}) \\ &= (-Ze) (E) \quad \dots (2)\end{aligned}$$

$$\text{Coulomb force} = \frac{\left[\text{Charge of the positive nucleus} \right] \times \left[\text{Total negative charges enclosed in the sphere} \right]}{4\pi\epsilon_0 d^2} \quad \dots (3)$$

$$\begin{aligned}\left. \begin{array}{l} \text{The total number of} \\ \text{negative charges enclosed} \\ \text{in the sphere of radius 'd'} \end{array} \right\} &= \left[\text{Charge density of electrons} \right] \times \left[\text{volume of the sphere} \right] \\ &= \left[\frac{-3Ze}{4\pi R^3} \right] \left[\frac{4\pi d^3}{3} \right] \\ &= \frac{-Zed^3}{R^3} \quad \dots (4)\end{aligned}$$

Substitute equation (4) in equation (3),

$$\begin{aligned}\text{Coulomb force} &= \frac{Ze \times \left[\frac{-Zed^3}{R^3} \right]}{4\pi\epsilon_0 d^2} \\ &= \frac{-Z^2 e^2 d}{4\pi\epsilon_0 R^3} \\ \text{Coulomb force} &= \frac{-Z^2 e^2 d}{4\pi\epsilon_0 R^3} \quad \dots (5)\end{aligned}$$

At equilibrium, coulomb force and Lorentz force must be equal and opposite.

$\text{Lorentz force (eqn (2))} = \text{Coulomb force (eqn (5))}$

$$(-Ze)(E) = \frac{-Z^2 e^2 d}{4\pi\epsilon_0 R^3}$$

$$E = \frac{Zed}{4\pi\epsilon_0 R^3}$$

$$d = \frac{4\pi\epsilon_0 R^3 E}{Ze} \quad \dots (6)$$

The displacement (d) of electron cloud is proportional to applied field (E).

Now the two electric charges $+Ze$ and $-Ze$ are displaced by a distance d under the influence of the field and form an induced dipole moment which is given by

Induced dipole moment (μ_e) = Charge of the electron $\times$ displacement

$$\mu_e = Ze d$$

Substituting the value of ' d ' from eqn (5)

$$\mu_e = \frac{Ze 4\pi\epsilon_0 R^3 E}{Ze}$$

$$\mu_e = 4\pi\epsilon_0 R^3 E$$

$$\mu_e \propto E$$

$$\mu_e = \alpha_e E$$

Where $\alpha_e = 4\pi\epsilon_0 R^3$ is called electronic polarization which is proportional to volume of the atom.

Conclusion:

- i) Electronic polarization is independent of temperature.
- ii) Electronic polarization occurs in all dielectric materials.
- iii) It is proportional to the volume of atoms in the material.

4.3.3 Electronic polarization in terms of ϵ_0 and ϵ_r

P_e is the total number of dipoles per unit volume [$V=1$]

$$P_e = \frac{N\mu_e}{V} = N\mu_e$$

Where N is the number of atoms per m^3

But $\mu_e = \alpha_e E$

$$P_e = N\alpha_e E$$

We know that, $P_e = \epsilon_0 E (\epsilon_r - 1)$

$$N\alpha_e E = \epsilon_0 E (\epsilon_r - 1)$$

$$\alpha_e = \frac{\epsilon_0 (\epsilon_r - 1)}{N}$$

This is called electronic Polarisability

4.3.4 Ionic Polarization

Ionic Polarization which arises due to the displacement of cations (+ve) and anions (-ve) in opposite directions and occurs in ionic solids in the presence of electric field, is called ionic polarization.

Example : NaCl, KCl crystal.

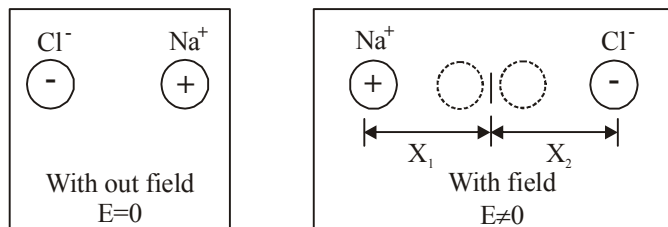


Fig. 4.3 Ionic polarization

Let us assume that there is one cation and one anion present in each unit cell of the ionic crystal (NaCl). When an electric field is applied let X_1 and X_2 be the distance to which positive and negative ions move from their equilibrium positions.

The Resultant dipole moment per unit volume due to ionic displacement is given by,

Dipole moment $\mu_i = \text{Charge} \times \text{Displacement}$

$$\mu_i = e[x_1 + x_2] \quad \dots (1)$$

When the field is applied, the restoring force produced is proportional to the displacement.

For +ve ion Restoring force $F \propto x_1$

$$F = \beta_1 x_1 \quad \dots (2)$$

For –ve ion Restoring force $F \propto x_2$

$$F = \beta_2 x_2 \quad \dots (3)$$

Here β_1 and β_2 are restoring force constants, which depends on the mass of the ions and angular frequency of the molecule in which ions are present.

If m is mass of positive ion, M is mass of negative ion, and ω_0 is angular frequency we can write,

$$\beta_1 = m \omega_0^2 \quad \dots (4)$$

$$\beta_2 = M \omega_0^2 \quad \dots (5)$$

Equation (2) and (3) can be written as,

$$F = m \omega_0^2 x_1 \quad \dots (6)$$

$$F = M \omega_0^2 x_2 \quad \dots (7)$$

We know, $F = eE \quad \dots (8)$

Equation equations (6), (7) with (8)

$$m \omega_0^2 x_1 = eE$$

$$x_1 = \frac{eE}{m \omega_0^2} \quad \dots (9)$$

$$M \omega_0^2 x_2 = eE$$

$$x_2 = \frac{eE}{M \omega_0^2} \quad \dots (10)$$

$$x_1 + x_2 = \frac{eE}{\omega_0^2} \left[\frac{1}{m} + \frac{1}{M} \right] \quad \dots (11)$$

Substitute equation (11) in equation (1)

$$\mu_i = \frac{e^2 E}{\omega_0^2} \left[\frac{1}{m} + \frac{1}{M} \right] \quad \dots (12)$$

We know that $\mu_i = \alpha_i E$

where α_i is ionic polarisability

$$\boxed{\alpha_i = \frac{e^2}{\omega_0^2} \left[\frac{1}{m} + \frac{1}{M} \right]}$$

Conclusion:

- i) Ionic polarisability (α_i) is inversely proportional to the square of angular frequency of the ionic molecule.
- ii) It is independent of temperature.
- iii) It is directly proportional to its reduced mass given by $\left[\frac{1}{m} + \frac{1}{M} \right]$

4.3.5 Orientation Polarization

The orientation polarization is due to the existence of a permanent dipole moment (polar molecule) in the dielectric medium. Polar molecules have permanent dipole even in the absence of an electric field.

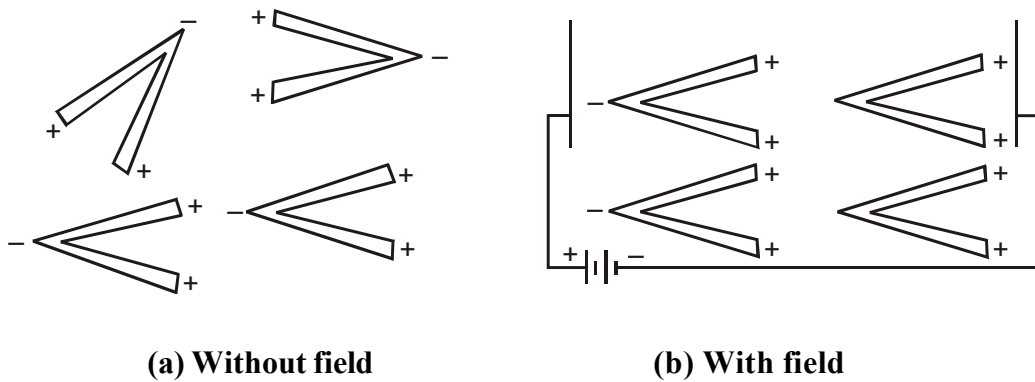


Fig.4.4 Orientation polarization

When an electric field is applied on the dielectric medium with polar molecules, the electric field tries to align these dipoles along its field direction as shown in figure. Due to this there is a resultant dipole moment in that material and this process is called orientation polarization.

This polarization depends on temperature. When the temperature is increased, the dipoles are aligned in random directions.

$$\text{Orientation polarization } P_o = \frac{N\mu^2 E}{3KT}$$

Where N - Number of atoms.

$$\text{Orientation polarization } P_o = N\alpha_o E$$

$$\alpha_o = \frac{P_o}{NE}$$

Here $\alpha_o \propto \frac{1}{T}$ i.e., α_o is inversely proportional to temperature of the material.

4.3.6 Space Charge Polarization

The space charge polarization occurs due to the diffusion of ions along the field direction giving rise to redistribution of charges in the dielectric.

Normally this type of polarization occurs in ferrites and semiconductors and it is very small when compared to other types of polarization.

4.4 FREQUENCY AND TEMPERATURE DEPENDENCE OF POLARIZATION MECHANISM

4.4.1 Frequency dependence

On application of an alternating field across the material, the polarization occurs as function of time

$$\text{i.e., } P(t) = P_m [1 - \exp[-t / t_r]]$$

Where P_m is the maximum polarization attained due to applied field and t is the relaxation time. Which is the time taken for a polarization process to reach 0.63 of the maximum value. The relaxation times are different for different kinds of polarization mechanisms.

1. **Electronic polarization** is very fast and is completed at any instant of time even when the frequency of the voltage is very high in the optical range (10^{15} Hz). Thus it occurs at all frequencies.
2. **Ionic Polarization** is slower and the ions do not respond when the voltage corresponds to visible optical frequencies, i.e., the electric field changes in polarity at very fast, so that the ions are not able to reorient themselves due up to the field. So the ionic polarization does not occur at visible optical frequencies. It occurs only at frequencies less than 10^{13} Hz.
3. **Orientation Polarization** is even slower than ionic polarization and occurs only at electrical frequencies (audio and radio frequencies 10^6 Hz).
4. **Space-charge polarization** is the slowest process because the ions have to diffuse (jump) over several inter atomic distances. This occurs at very low frequencies of 50 - 60 Hz (power frequencies).

Thus at low frequencies all the four polarizations will occur and the total polarization is very high, but at high frequencies, the value of the total polarization is very small. The following graphs show the frequency dependence of polarization mechanism and the corresponding power losses at those frequencies.

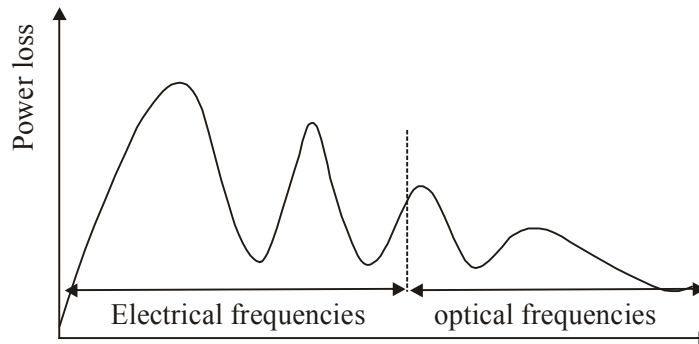
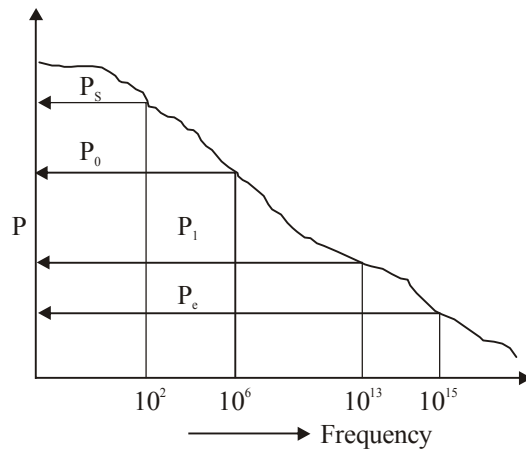


Fig.4.6 Polarization Mechanisms

4.4.2 Temperature dependence

Electronic and ionic polarizations are independent of temperature and the orientation and space charge polarizations are dependent of temperature. Orientation polarization is inversely proportional to the temperature.

Orientation polarization decreases when temperature increases. Because the random nature decreases the tendency of permanent dipoles to align along the field direction. Thus the dielectric constant increases.

Space charge polarization is directly proportional to the temperature. The space charge polarization increases with increase the temperature. It is because of the fact that the thermal energy helps to overcome the activation barrier and the ions diffuse easily, this results in decrease of dielectric constant.

4.5 COMPARISION OF TYPES OF POLARISATION

<i>S. N</i>	<i>Factor</i>	<i>P_e</i>	<i>P_i</i>	<i>P_o</i>	<i>P_s</i>
1	Definition	Electron clouds are shifted from nucleus	Cations and anions are shifted	Regular alignment of random molecules takes place.	Ion diffusion takes place.
2	Examples	Inert gases	Ionic crystals	Alcohol, methane, CH ₃ Cl	Semi conductors, Ferrites.
3	Temperature dependence	Independent	Independent	Dependent	Dependent
4	Relaxation time	Very fast	Slow	Slower	Slowest
5	Power loss	Low	High	Higher	Highest
6	Frequency range	10 ¹⁵ Hz	10 ¹³ Hz	10 ⁶ Hz	10 ⁵ Hz
7	Polarisability	$\alpha_e = \frac{\epsilon_0 (\epsilon_r - 1)}{N}$	$\alpha_i = \frac{e^2}{\omega_0^2} \left[\frac{1}{m} + \frac{1}{M} \right]$	$\alpha_o = \frac{\mu^2}{3KT}$	α_s is negligible

4.6 INTERNAL FIELD (OR) LOCAL FIELD

When a dielectric material is placed in an external electric field, it exerts a dipole moment in it. Therefore two fields are exerted.

1. Due to external field.
2. Due to dipole moment

This long range of coulomb forces which is created due to the dipoles is called as **Internal field (or) local field**. This field is responsible for polarizing the individual atoms (or) molecules.

Lorentz method to find the internal field

The dielectric material which is placed in between two plates of a parallel plate capacitor is uniformly polarized as shown in Figure.

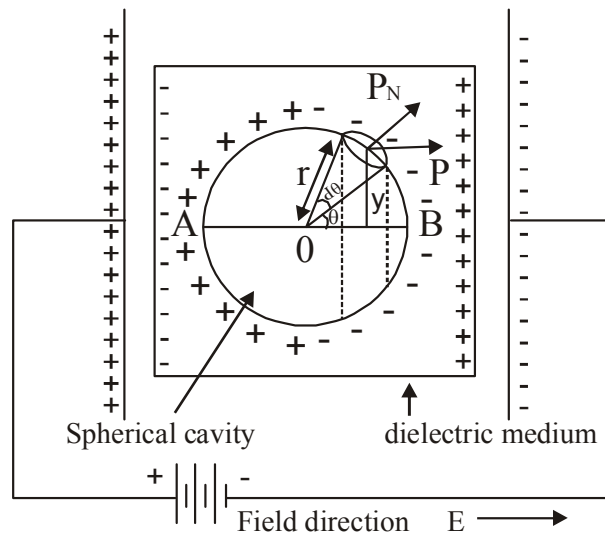


Fig.4.7 Calculation of local field for a cubic structure

Assume an imaginary small spherical cavity around an atom for which the internal field must be calculated at its centre.

The internal field E_{int} at the atom site (0) can be considered to consist of components namely E_1 , E_2 , E_3 and E_4 .

$$\text{i.e.,} \quad E_{\text{int}} = E_1 + E_2 + E_3 + E_4 \quad \dots (1)$$

E_1 – Electric field due to charges on this plates of the capacitor.

E_2 – Electric field due to the polarised charges (induced charges) on the plane surface of the dielectric.

E_3 – Electric field due to polarised charges induced on the surface of the imaginary spherical cavity.

E_4 – Electric field due to permanent dipoles of the atoms inside the spherical cavity considered.

Macroscopically, we can take $E = E_1 + E_2$ i.e., the field externally applied (E_1) and the field induced on the plane surface of the dielectric (E_2) can be considered as a single field (E).

If we consider the dielectric which is highly symmetric, then the field due to the dipoles present inside the imaginary cavity will cancel each other. Therefore, the electric field due to permanent dipoles $E_4 = 0$.

The equation (1) can be written as

$$E_{\text{int}} = E + E_3 \quad \dots (2)$$

Determination of E_3

Let us consider a small area ds on the surface of the spherical cavity. This small area makes an angle $d\theta$ at the center at an angle θ with the direction of the field E .

The polarization P will be parallel to E . P_N is the component of polarization perpendicular to the area ds , and q is the charge on the area ds .

Polarization is also defined as the surface charges per unit area.

Hence
$$P_N = P \cos \theta = \frac{q}{ds}$$

Charge on ds , $q = P \cos \theta ds \quad \dots (3)$

Electric field intensity at 'C' due to charge q (Coulomb's law)

$$E = \frac{q}{4\pi\epsilon_0 r^2} \quad \dots (4)$$

Substitute the equation (3) in (4) we get

$$E = \frac{P \cos \theta ds}{4\pi\epsilon_0 r^2} \quad \dots (5)$$

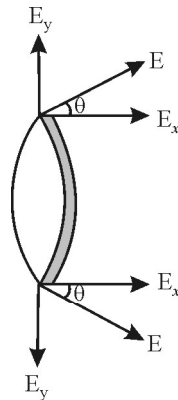


Fig. 4.8

This field intensity is along the radius 'r' and it can be resolved into two components as shown in Figure.

The component of intensity *parallel* to the field direction

$$E_x = E \cos \theta \quad \dots (6)$$

Substitute the equation (5) in (6) we get

$$E_x = \frac{P \cos \theta \, ds \cos \theta}{4\pi\epsilon_0 r^2}$$

$$\text{i.e.,} \quad E_x = \frac{P \cos^2 \theta \, ds}{4\pi\epsilon_0 r^2} \quad \dots (7)$$

The component of intensity *perpendicular* to the field direction

$$E_y = E \sin \theta \quad \dots (8)$$

$$\therefore E_y = \frac{P \cos \theta \sin \theta \, ds}{4\pi \epsilon_0 r^2}$$

Since the perpendicular components are in opposite direction figure, they cancel out each other. So the parallel components alone are taken into consideration.

Consider a ring area ds which is obtained by revolving ds about AB figure

Ring area $\rightarrow ds = \text{Circumference} \times \text{thickness}$

$$\begin{aligned} &= 2\pi y \times r d\theta & \therefore \sin \theta &= \frac{y}{r} \\ &= 2\pi r \sin \theta \times r d\theta & y &= r \sin \theta \end{aligned}$$

$$\text{i.e.,} \quad ds = 2\pi r^2 \sin \theta \, d\theta \quad \dots (9)$$

Electric field intensity due to the elemental ring dA

$$= \frac{P \cos^2 \theta \, ds}{4\pi\epsilon_0 r^2} \quad \dots (10)$$

Substituting equation (9) in (7) we get

$$\begin{aligned} &= \frac{P \cos^2 \theta}{4\pi \epsilon_0 r^2} \times 2\pi r^2 \sin \theta \, d\theta \\ &= \frac{P \cos^2 \theta \sin \theta \, d\theta}{2\epsilon_0} \quad \dots (11) \end{aligned}$$

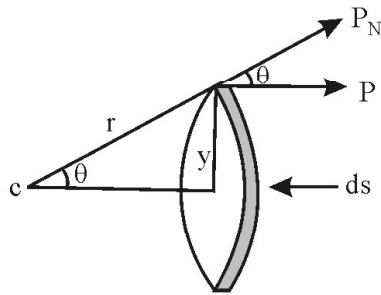


Fig.4.9

Electric field intensity due to charge present in the whole sphere is obtained by integrating equation (11) within the limits 0 to π . This field is taken as E_3 .

$$E_3 = \int_0^\pi \frac{\pi P \cos^2 \theta \sin \theta d\theta}{2\epsilon_0}$$

$$E_3 = \frac{P}{2\epsilon_0} \int_0^\pi \cos^2 \theta \sin \theta d\theta \quad \therefore \int_0^\pi \cos^2 \theta \sin \theta d\theta = \frac{2}{3}$$

$$E_3 = \frac{P}{2\epsilon_0} \times \frac{2}{3}$$

$$E_3 = \frac{P}{3\epsilon_0} \quad \dots (12)$$

Substituting equation (12) in (2), we get

$$E_{\text{int}} = E + \frac{P}{3\epsilon_0} \quad \dots (13)$$

Where E_{int} is called internal field or Lorentz field.

This equation (13) shows that E_{int} is different from E . The local intensity E_{int} is larger than the macroscopic intensity E . So the molecules are more effectively polarised.

4.6.1 Clausius Mosotti Equations

If N be the number of molecules per unit volume and α the molecular polarizability then total polarization $P = N \alpha E_{\text{int}}$

$$E_{\text{int}} = \frac{P}{N\alpha} \quad \dots (14)$$

Further, we know that

$$\begin{aligned} D &= \epsilon E = \epsilon_0 E + P \\ (\epsilon - \epsilon_0)E &= P \\ E &= \frac{P}{(\epsilon - \epsilon_0)} \quad \dots (15) \end{aligned}$$

Lorentz field is given by

$$E_{\text{int}} = E + \frac{P}{3\epsilon_0} \quad \dots (16)$$

Substituting the eqn (14) & eqn (15) in eqn. (16) we get

$$\begin{aligned} E_{\text{int}} &= \frac{P}{\epsilon - \epsilon_0} + \frac{P}{3\epsilon_0} \\ &= P \left[\frac{3\epsilon_0 + \epsilon - \epsilon_0}{3\epsilon_0(\epsilon - \epsilon_0)} \right] \\ E_{\text{int}} &= \frac{P}{3\epsilon_0} \left[\frac{\epsilon + 2\epsilon_0}{\epsilon - \epsilon_0} \right] \quad \dots (17) \end{aligned}$$

Comparing equation (14) and (17), we get

$$\begin{aligned} \frac{P}{N\alpha} &= \frac{P}{3\epsilon_0} \left[\frac{\epsilon + 2\epsilon_0}{\epsilon - \epsilon_0} \right] \\ \frac{N\alpha}{3\epsilon_0} &= \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \end{aligned}$$

$$= \frac{(\epsilon / \epsilon_0) - 1}{(\epsilon / \epsilon_0) + 2}$$

$$\frac{N\alpha}{3\epsilon_0} = \frac{\epsilon_r - 1}{\epsilon_r + 2} \quad \left[\because \epsilon_r = \frac{\epsilon}{\epsilon_0} \right] \quad \dots (14)$$

The above equation is Clausius – Mosotti relation, which relates the dielectric constant of the material and polarisability. Thus, it relates macroscopic quantity dielectric constant with microscopic quantity polarisability.

4.7 DIELECTRIC LOSS

When a dielectric material is subjected to an alternating electric field, some amount of energy is absorbed by the material and is dissipated in the form heat. This loss of energy is called Dielectric loss.

The dielectric loss depends on the type of dielectric medium and the following factors.

- | | |
|----------------------|---------------|
| i) Temperature | ii) Humidity |
| iii) Applied voltage | iv) Frequency |

Theory

In an ideal dielectric material, the current leads the voltage by 90° . (i.e. no loss of energy) i.e. $\theta = 90^\circ$

But in a commercial dielectric material, the current leads the voltage by less than 90° . (i.e., there is loss of energy), i.e., the angle $\phi = 90^\circ - \theta$ is known as the dielectric loss angle.

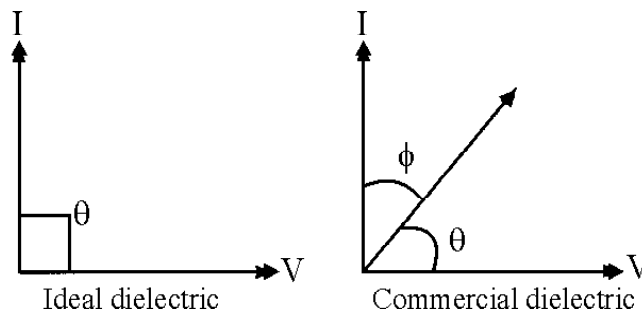


Fig. 4.10 Dielectric Loss

Consider a dielectric having a capacitance of C and having a voltage V applied to it at a frequency f .

The power loss in the dielectric material is given by $P = V I \cos \phi$

We know that $V = IR$ $\therefore I = \frac{V}{R}$

Frequency $f = \frac{1}{2\pi RC}$

If the capacitance reactance is X_C , then we can write

$$I = \frac{V}{X_C}$$

Frequency $f = \frac{1}{2\pi X_C C}$

Where, $X_C = \frac{1}{2\pi f C}$

$$P = \frac{V^2}{2\pi f C} \cos(90^\circ - \theta)$$

But, $[\phi = 90 - \theta]$ $P = V^2 2\pi f C \cos(90^\circ - \theta)$

$$P = V^2 2\pi f C \sin \theta$$

In most of the dielectrics, the angle θ is very small $\therefore \sin \theta = \tan \theta$

$$P = V^2 2\pi f C \tan \theta$$

Where $\tan \theta$ is called the power factor of the dielectric.

It is noted that the power loss will depend on $\tan \theta$ as long as other factors like voltage, frequency.

Conclusions

1. The dielectric loss increases with increase of frequency, applied voltage, temperature, and humidity.
2. The dielectric loss is maximum, when the relaxation time of a polarization process matches the period of the applied AC voltage.
3. The dielectric loss at radio frequency is high due to diffusion of ions.
4. The dielectric losses in the optical region, associated with the electrons are referred to as optical absorption. This absorption leads to color of materials.
5. The dielectric losses in the infrared region associated with the ionic vibrations are referred to as Infrared absorption.

4.8 DIELECTRIC BREAKDOWN

Definition

When a dielectric material loses its property and permits the flow of a large current, it is said to be dielectric breakdown.

4.8.1 Types of dielectric breakdown

1. Intrinsic breakdown
2. Thermal breakdown.
3. Electrochemical breakdown.
4. Defect breakdown.
5. Discharge breakdown.

1. Intrinsic Breakdown

When a dielectric material is subjected to large electric field, a large number of electrons are transferred from the valence band to the conduction band. Thus the dielectric material loses its insulating property and becomes a conductor. This is known as intrinsic breakdown.

Further the conducting electrons may collide with the atoms and release some of the valence electrons. These electrons may collide with some more atoms and may release more electrons. This becomes a chain process resulting in a large current. It is known as avalanche breakdown.

Characteristics

- i) This type of breakdown occurs at room temperature and low temperatures.
- ii) This requires relatively large electric fields
- iii) This kind of breakdown occurs in the materials.

2. Thermal breakdown

Thermal breakdown occurs in dielectric when the rate of heat generation is greater than the rate of heat dissipation.

When a dielectric is subjected to an electric field, heat is generated. The generated heat is dissipated by the dielectric material. When the amount of heat generation is higher than the amount of heat dissipation, the temperature inside the dielectric increases and this causes the breakdown called thermal breakdown.

Characteristics:

- i) This can occur only at high temperatures.
- ii) The Thermal breakdown time is of the order of few milliseconds.
- iii) This requires moderate electric fields.
- iv) It depends on size and shape of dielectric material.
- v) Since the dielectric loss is directly proportional to frequency, the electric field strength to create this dielectric breakdown will be smaller for a.c fields and higher d.c fields.

3. Discharge breakdown

This type of breakdown occurs when the insulator contains occluded gas bubbles. When the dielectric is subjected to an electric field, the gas present in the material will be easily ionized than the solids.

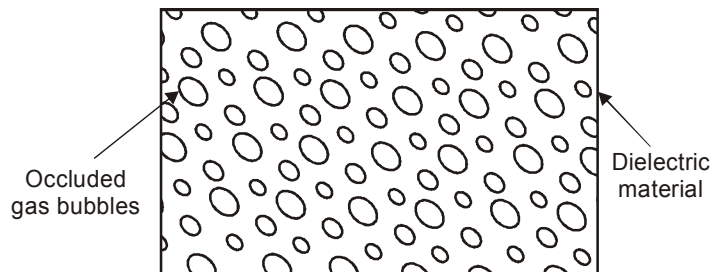


Fig. 4.11 Discharge breakdown

The ionized gas particles bombarded with the solid dielectric and produce large ionization current called discharge breakdown.

Characteristics

- i) This is possible at low voltages where, there are large number of occluded gas bubbles is present in the material.
- ii) The life time of the material depends upon the number of discharge taking place inside the material.
- iii) This occurs due to presence of gas bubbles.

4. Electrochemical breakdown

When temperature increases, mobility of ions increases and hence leakage current also increases. This decreases the insulation resistance and finally creates a dielectric breakdown. Hence this type of breakdown is called electrochemical breakdown.

Characteristics

- i) This is used to determine the leakage current, density of ions and dipoles inside the material.
- ii) They are accelerated by high temperatures
- iii) This occurs only at low temperature
- iv) This type of breakdown occurs even in absence of electric field also.

5. Defect breakdown

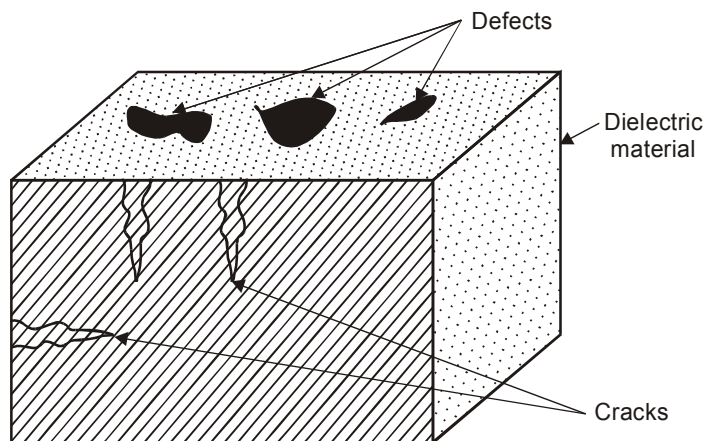


Fig. 4.12 Defect breakdown

If the surface of the dielectric material has defects such as cracks, pores, etc. Moisture and other impurities can fill at these places leading to breakdown, this type of breakdown called Defect breakdown.

Remedies to Avoid Breakdown Mechanisms

To avoid breakdown, the dielectric material should have the following properties

1. It should have high resistivity.
2. It must have low dielectric loss and less density.
3. It should have sufficient mechanical strength.
4. Thermal expansion should be small.
5. It should be fire proof.
6. It must in pure form and not have any defects.

4.9 FERRO – ELECTRICITY AND ITS APPLICATIONS

Ferro electricity refers to the creation of large induced dipole moment in a weak electric field as well as the existence of electric polarization even in the absence of an applied electric field.

Crystalline dielectric materials which possess a permanent electric polarization are called ferroelectric materials have electric dipole moment even in the absence of any field. Normally they are anisotropic crystals which exhibit spontaneous polarization.

Examples:

1. Barium Titanate (BaTiO_3)
2. Potassium Dihydrogen Phosphate (KDP)
3. Ammonium Dihydrogen Phosphate ($\text{NH}_4\text{H}_2\text{PO}_4$)
4. Lithium Niobate (LiNbO_3)

4.9.1 Properties of Ferroelectric Materials

1. Ferroelectric materials exhibit spontaneous polarization i.e., they are polarized even in the absence of electric field.
2. They exhibit dielectric hysteresis. The lagging of polarization behind the applied electric field is called dielectric hysteresis.

3. Above ferroelectric Curic temperature, ferroelectric material becomes paraelectric material.
4. They posses Ferro – electric domain structure.
5. They can be polarized even by very weak electric field
6. Ferroelectric materials exhibit piezoelectricity and Pyro electricity. Piezoelectricity means the creation of electric polarization by mechanical stress. Pyro electricity means the creation of electric polarization by thermal stress.

4.9.2 Hysteresis of Ferroelectric Materials

The ferro electrics are known as non –linear dielectrics. Such materials exhibit hysteresis curve similar to that of ferro magnetic materials.

The lagging of polarization ‘P’ behind the applied electric field E is called dielectric hysteresis.

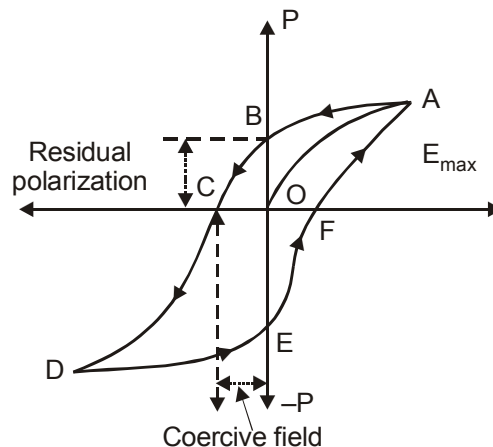


Fig 4.13 Dielectric Hysteresis in ferroelectric materials

When a ferro – electric material is subjected to external electric field (E) the polarization (P) increases with respect to the field applied and it reaches the maximum value ‘OA’.

If now the applied electric field is reduced, the polarization also decreases from a, and when E becomes zero a small amount of polarization exists in the material is called spontaneous (or) residual polarization.

In order to reduce the value of polarization to zero, a reversing electric field should be applied. This field is known as coercive field.

Thus the variation of P with respect to E traced along the closed path abcdefa in one full cycle of polarization and depolarization is called hysteresis or the hysteresis curve.

4.9.3 Application of Ferroelectric Materials

1. Ferroelectric materials are used to make pressure transducers, ultrasonic transducers, microphones and gas filters.
2. They are used as memory cores in computers.
3. They are used to measure and control temperature.
4. Ferroelectric ceramics are used as capacitors to store electrical energy
5. They are used to make very good infrared detectors.
6. Rochelle salt is used in devices like microphones, strain gauges, phonograph pickups and SONAR devices.
7. In optical communication, the ferroelectric crystals are used for optical modulation.
8. Ferro electric materials are used to produce ultrasonics
9. Electrets are also used to bond the fractured bones in human body.
10. They are used as frequency stabilizers and crystal controlled oscillators.

4.10 APPLICATIONS OF DIELECTRIC MATERIALS

The dielectric materials have three major applications.

1. It is used as a dielectric medium in capacitors.
2. It is used as insulating materials in transformers.
3. It is used in industries and dielectric heating.

4.10.1 Dielectrics in Capacitors

If dielectrics, to be used in capacitors, it should posse the following properties.

Properties

1. It must have high dielectric constant
2. It should posses high dielectric strength
3. It should also have low dielectric loss.

Applications

1. Thin sheets of papers filled with synthetic oils are used as dielectrics in the capacitors.
2. Tissue papers and poly propylene films filled with dielectrol are used in power capacitors.
3. Mica is used as dielectrics in discrete capacitors.
4. An electrolytic solution of sodium phosphare is used in wet type electrolytic capacitors.

4.10.2 Insulating materials in transformers

If dielectrics to act as insulating materials, it should possess the following properties.

Properties

1. It should have low dielectric constant
2. It should possess low dielectric loss
3. It must have high resistance
4. It must possess high dielectric strength
5. It should have adequate chemical stability
6. It must have high moisture resistance

Applications

1. PVC (Poly Vinyl Chloride) is used to manufacture pipes, batteries, cables etc.
2. Liquid dielectrics such as petroleum oils, silicone oils are widely used in transformers, circuit breakers etc.
3. Synthetic oils such as askarels, sovol, etc., are used as a coolant and insulant in high voltage transformers.
4. Gases such as nitrogen, sulphur hexa fluoride are used in x-ray tubes, switches, high voltage gas filled pressure cables, coolants respectively.

5. Industrial application

- a. Dielectrics possessing piezoelectric effect is used in gas lighters, microphones, phonographs.
- b. Dielectrics possessing inverse Piezo electric effect is used in quartz watchers, ultrasonic dryers, cleaning the semiconducting wafers, ultrasonic transducer etc.

6. Dielectric Heating

Dielectric heating is the process of heating the insulating materials at a very high voltage under suitable frequency at which dielectric loss becomes maximum, so that the dielectric loss will come out in the form of heat. Hence adequate heating was done at high voltages.

- a. Dielectric heating is the principle used in microwave oven.
- b. Dielectric heating is also used in the dehydration of food, tobacco etc.

SOLVED PROBLEMS

1. Find the capacitance of a layer of Al_2O_3 that is $0.5 \mu\text{m}$ thick and 2000 mm^2 of square area ($\epsilon_r = 8.854 \times 10^{-12} \text{ Fm}^{-1}$)

Solution:

Given data:

Permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12}$

Area of the capacitance $A = 2000 \text{ mm}^2$ (or) $2000 \times 10^{-6} \text{ m}^2$

Thickness $d = 0.5 \mu\text{m}$ (or) $0.5 \times 10^{-6} \text{ m}$

$$C = \frac{\epsilon_r \epsilon_0 A}{d}$$
$$= \frac{8.854 \times 10^{-12} \times 2000 \times 10^{-6}}{0.5 \times 10^{-6}}$$

$C = 3.5416 \times 10^{-8} \text{ F}$

2. A layer of porcelain is 80 mm long, 20 mm wide and $0.7 \mu\text{m}$ thick. Calculate its capacitance with $\epsilon_r = 6$

Solution:

Given data:

Permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12} \text{ F m}^{-1}$

Thickness $d = 0.7 \mu\text{m}$ (or) $0.7 \times 10^{-6} \text{ m}$

Area = $l \times b \times h$

$$C = \frac{\epsilon_r \epsilon_0 A}{d}$$
$$= \frac{8.854 \times 10^{-12} \times 6 \times 80 \times 20 \times 10^{-6}}{0.7 \times 10^{-6}}$$

$C = 1.21 \times 10^{-7} \text{ F}$

3. If an ionic crystal is subjected to an electric field of 1000 Vm^{-1} and the resulting polarization $4.3 \times 10^{-8} \text{ cm}^2$. Calculate the relative permittivity of NaCl.

Solution:

Given data:

$$\text{Electric field } E = 1000 \text{ Vm}^{-1} \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$$

$$P = 4.3 \times 10^{-8} \text{ cm}^2$$

$$P = \epsilon_0(\epsilon_r - 1)E$$

$$\epsilon_r = 1 + \frac{P}{\epsilon_0 E}$$

$$\epsilon_r = 1 + \frac{4.3 \times 10^{-8}}{(8.85 \times 10^{-12} \times 1000)}$$

$$\boxed{\epsilon_r = 5.86}$$

4. Calculate the electronic Polarization of argon atom. Given at $\epsilon_r = 1.0024$ at NTP and $N = 2.7 \times 10^{25} \text{ atom m}^{-3}$.

Solution :

$$\text{Relative permittivity } \epsilon_r = 1.0024 \text{ at NTP}$$

$$\text{Permittivity of free space } \epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$$

$$\text{Number of atoms } N = 2.7 \times 10^{25} \text{ atom / m}^3$$

$$\alpha_e = ?$$

$$P = N\alpha_e E$$

$$\alpha_e = \frac{P}{NE}$$

$$P = \epsilon_0(\epsilon_r - 1)E$$

$$\alpha_e = \frac{\epsilon_0(\epsilon_r - 1)E}{NE} = \frac{\epsilon_0(\epsilon_r - 1)}{N}$$

$$\alpha_e = \frac{8.85 \times 10^{-12} \times (1.0024 - 1)}{2.7 \times 10^{25}}$$

$$\boxed{\alpha_e = 7.9 \times 10^{-40} \text{ Fm}^{-2}}$$

5. A solid contains 5×10^{28} atoms/ m^3 each with a polarisability of $2 \times 10^{-40} \text{ F m}^2$. Assuming that the internal field is given by Lorentz formula. Calculate the ratio of internal field to the external field. $\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$.

(AU - MAY 2011)

Solution:

Given data:

Polarizability $\alpha = 2 \times 10^{-40} \text{ Fm}^2$

Number of atoms $N = 5 \times 10^{28} \text{ atoms / m}^3$

We know Lorentz field, in the internal field

$$E_i = E + \frac{P}{3\epsilon_0}$$

Also we know $P = N \alpha E_i$

Therefore $E_i = E + \frac{N \alpha E_i}{3\epsilon_0}$

$$E_i \left[1 - \frac{N \alpha}{3\epsilon_0} \right] = E$$

$$\frac{E_i}{E} = \frac{1}{\left(1 - \frac{N \alpha}{3\epsilon_0} \right)}$$

$$\frac{E_i}{E} = \left[\frac{1}{1 - \left(5 \times 10^{28} \times 2 \times 10^{-40} / 3 \times 8.85 \times 10^{-12} \right)} \right]$$

$$\boxed{\frac{E_i}{E} = 1.604}$$

6. Calculate the polarization produced in dielectric medium of dielectric constant 6 when it is subjected to an electric field of 100 Vm^{-1} . ($\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$) (AU - JUN 2009, 2012)

Soution:

Given data:

$$\text{Dielectic constant } \epsilon_r = 6$$

$$\text{Permitivity of free space } \epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$$

$$\text{Electric field } E = 100 \text{ Vm}^{-1}$$

$$\begin{aligned} \text{Polarization } P &= \epsilon_0 (\epsilon_r - 1) E \\ &= 8.854 \times 10^{-12} (6-1) 100 \end{aligned}$$

$$P = 4.425 \times 10^{-9} \text{ Vm}^{-2} \text{ F}$$

7. Calculate the electronic polarizability of neon. The radius of neon atom is 0.158 nm . ($\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$) (AU - DEC 2010)

Solution:

Given data:

$$\text{Radius of neon atom } R = 0.158 \text{ nm} \quad (\text{or})$$

$$= 0.158 \times 10^{-9} \text{ m}$$

$$\text{Permitivity of free space } \epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$$

$$\text{Electronic polarizability} \quad \alpha_e = 4\pi\epsilon_0 R^3$$

$$\begin{aligned} \alpha_e &= 4(3.14) (8.854 \times 10^{-12}) (0.158 \times 10^{-9})^3 \\ &= 4.386 \times 10^{-40} \text{ Fm}^2 \end{aligned}$$

$$\text{Electronic polarizability}$$

$$\alpha_e = 4.386 \times 10^{-40} \text{ Fm}^2$$

8. The dielectric constant of a helium gas at NTP is 1.0000684. Calculate the electron polarizability of helium atoms if the gas contains 2.7×10^{26} atoms/m³ and hence calculate the radius of helium atom ($\epsilon_0 = 8.854 \times 10^{-12}$ Fm⁻¹)

Solution:

Given data:

$$\text{Relative permittivity } \epsilon_r = 1.0000684$$

$$\text{No. of atoms in the gas } N = 2.7 \times 10^{26} \text{ atoms/m}^3$$

$$\text{Permittivity of free space } \epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$$

i) We know polarization

$$P = \epsilon_0 (\epsilon_r - 1) E$$

$$\text{and } P = N\alpha_e E$$

From the above two equations, we can write

$$N\alpha_e = \epsilon_0 (\epsilon_r - 1)$$

$$\alpha_e = \frac{\epsilon_0 (\epsilon_r - 1)}{N}$$

$$= \frac{8.854 \times 10^{-12} (1.0000684 - 1)}{2.7 \times 10^{26}}$$

$$\alpha_e = 2.242 \times 10^{-42} \text{ Fm}^2$$

ii) Electronic polarizability

$$\alpha_e = 4\pi\epsilon_0 R^3$$

$$R = \frac{\alpha_e}{4\pi\epsilon_0}$$

$$= \frac{2.242 \times 10^{-42}}{4 \times 3.14 \times 8.854 \times 10^{-12}}$$

Radius of helium atom

$$R = 0.0201 \times 10^{-30} \text{ m}$$

SHORT QUESTIONS WITH ANSWERS

1. What is a dielectric material? State their property.

Dielectrics are the insulating materials having electric dipole moment permanently or temporarily by applying the electric field. These are mainly used to store electrical energy and used as electrical insulators. All dielectrics are electrical insulators. But all electrical insulators need not to be dielectrics. Dielectrics are non-metallic materials of high specific resistance and have negative temperature coefficient of resistance.

2. Define electric flux density (D) or Electric Displacement Vector.

The number of electric lines passing through the unit area of cross section.

$$D = [\phi / A]$$

Unit: Coulomb / m²

3. Define Permittivity

It is the ratio of electric displacement vector in a dielectric medium to the applied electric field strength. $\epsilon = D / E$

4. Define Dielectric constant or Relative Permittivity (ϵ_r)

(AU - Nov 2010, June 2010)

It is the ratio of permittivity of the medium to the permittivity of the free space.

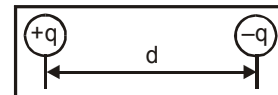
$$\epsilon_r = [\epsilon / \epsilon_0] \text{ where } \epsilon_0 = 8.854 \times 10^{-12} \text{ F/m.}$$

5. Define Dipole moment (μ)

Dipole moment is defined as the product of charge and distance.

$$\text{ie., } \mu = q \times d$$

Unit : Coulomb meter.



6. Define Polarization, Polarization vector and Polarisability

Polarization

The separation of negative and positive charges is called polarization. i.e., the process of producing electric dipoles by an electric field is called polarization.

$$\mu = \alpha E$$

where α = constant of proportionality called polarisability.

Polarization vector

If μ is the average dipole moment per molecule and N is the number of molecules per unit volume then the polarization of the solid is given by the polarization vector P and it can be written as

$$P = \frac{N\mu}{V} = N\mu \text{ [for unit volume } V=1]$$

The polarization vector is the dipole moment per unit volume of the dielectric material.

Polarisability

The polarization depends on electric field. $P \propto E$, $P = \alpha E$, $\alpha = P / E$ the proportionality constant is called as polarisability.

7. Define Electronic Polarization. (AU - April 2003, Nov 2003)

Electronic polarization occurs due to the displacement of positively charged nucleus and negatively charged electron in opposite directions by an external electric field. It creates a dipole moment in the dielectric. This is called electronic polarization.

$$\alpha_e = 4\pi\epsilon_0 R^3$$

8. What is ionic polarization. (AU - May 2004, Nov 2010, June 2010, Dec 2009)

The displacement of cations (+ve) and anions (-ve) in opposite directions is called ionic polarization. It occurs in ionic solids in the presence of electric field

$$\alpha_i = \frac{e^2}{\omega_0^2} \left(\frac{1}{m} + \frac{1}{M} \right)$$

9. Define orientation polarization. (AU - May 2005, Dec 2009)

When an electric field is applied on the dielectric medium with polar molecules, the electric field tries to align these dipoles along its field direction, due to that there is a resultant dipole moment in that material and this process is called orientation polarization.

$$\alpha_o = \frac{\mu^2}{3KT}$$

10. What is Space charge polarization?

(AU - May 2002)

The space charge polarization occurs due to the diffusion of ions along the field direction giving rise to redistribution of charge in the dielectrics. Normally this type of polarization occurs in ferrites and semiconductors and it is very small when compared to other types of polarization.

11. What are polar and non-polar molecules?

Molecules which are having permanent dipole moment even in the absence of an applied field are called polar molecules. Example: H_2O , HC , CO .

Molecules which do not have permanent dipole moment, but they have induced dipole moment in the presence of applied electric field are called non - polar molecules.

Example: O_2 , H_2 , N_2

12. Define Local or internal or Lorentz field.

(AU - Nov 2003, Dec 2003)

In a dielectric material, the field acting at the location of an atom is called local field or internal field $[E_i]$

The internal field E must be equal to the sum of applied field and the field due to location of the atom by the dipoles of all other atoms

ie, $E_i = E + \text{The field due to all other dipoles.}$

13. Define Dielectric Loss and Dielectric breakdown.

(AU - Nov 2010, May 2011, June 2013)

When a dielectric material is subjected to an alternating electric field, some amount of energy is absorbed by the material and is dissipated in the form heat. This loss of energy is called Dielectric loss.

When a dielectric material loses its property and permits the flow of a large current, it is said to be dielectric breakdown.

14. What is optical absorption and infrared absorption in a dielectric?

The dielectric losses in the optical region, associated with the electrons are referred to as optical absorption. This absorption leads to color of materials. The dielectric losses in the infra-red region, associated with the ionic vibrations are referred to as Infrared absorption.

15. Define Dielectric breakdown and what are its types?

(AU - April 2005, June 2009, Dec 2012)

When a dielectric material loses its property and permits the flow of a large current, it is known as dielectric breakdown.

Types

- 1) Intrinsic breakdown.
- 2) Thermal breakdown.
- 3) Electrochemical breakdown.
- 4) Defect breakdown.
- 5) Discharge breakdown.

16. What are Ferro electric materials?

(AU - June 2009, 2012)

Materials which exhibit electric polarization even in the absence of electric field are known as Ferro electric materials.

Crystalline dielectric materials which possess a permanent electric polarization are called ferroelectric materials. They have electric dipole moment even in the absence of any field. Normally they are anisotropic crystals which exhibit spontaneous polarization.

Examples : Barium Titanate [BaTiO_3], Potassium Dihydrogen Phosphate [KH_2PO_4], Lithium Niobate [LiNbO_3] and Rochelle salt.

17. What are the differences between polar and non - polar molecules?

(AU - Nov 2003)

S.No	Polar molecules	Non - Polar molecules
1.	These molecules have permanent dipole moment even in the absence of an applied field.	These molecules do not have permanent dipole moment
2.	The polarization of polar molecules is highly temperature dependent.	The polarization in these type of molecules is independent of temperature.
3.	There is absorption or emission in the infrared range for these molecules Example: H_2O , HCl , CO	There is no absorption or emission in infrared range for these molecules Example: O_2 , H_2 , N_2

18. Compare active and passive dielectrics.

(AU - April 2003, May 2007, June 2009)

<i>S. No.</i>	<i>Active dielectrics</i>	<i>Passive dielectrics</i>
1.	Dielectrics which can easily adapt to store the electrical energy in it is called active dielectrics.	Dielectrics which resulting the flow of electrical energy in it are called passive dielectrics.
2.	They are used in the production of ultrasonics.	They are used in production of sheets.
3.	Example : Piezo electrics, Ferro electrics	Example : Glass, mica and Plastic

19. What are the applications of ferroelectric materials? (AU - Dec 2009)

1. Ferroelectric materials are used to make pressure transducers, ultrasonic transducers, microphones and gas filters.
2. They are used as memory cores in computers.
3. They are used to measure and control temperature.
4. Ferroelectric ceramics are used as capacitors to store electrical energy
5. They are used to make very good infrared detectors.
6. Rochelle salt is used in devices like microphones, strain gauges, phonograph pickups and SONAR devices.
7. In optical communication, the ferroelectric crystals are used for optical modulation.
8. Ferro electric materials are used to produce ultrasonics
9. Electrets are also used to bond the fractured bones in human body.
10. They are used as frequency stabilizers and crystal controlled oscillators.

PART - B QUESTIONS

1. What is meant by polarization in dielectrics? Derive the relation between the dielectric constant and atomic polarisability.
2. Explain the different types of polarization mechanisms in a dielectric.
3. Explain the frequency and temperature dependence of various polarizations.
4. What is meant by local field? Derive an expression for Lorentz field in a dielectric material and hence deduct Clausius - Mosotti relation.
5. What is dielectric loss. Derive an expression for power loss in a dielectric.
6. Discuss in detail the various dielectric breakdown mechanisms.

ASSIGNMENT PROBLEMS

1. The electronic polarisability is $0.18 \times 10^{-40} \text{ fm}^2$. Find the relative dielectric constant at 0°C and 1 atmospheric pressure. **(Ans: 1.000055)**
2. A capacitor has capacitance of 0.019 F when uses wax paper ($\epsilon_r = 1.85$) between the electrodes of Aluminum foil. The wax paper is to be replaced by plastic film ($\epsilon_r = 2.15$) of same dimensions. Taking other factors being equal, obtain the change (increase/decrease) in capacitance. **(Ans: Increase of 0.0117 F)**
3. A capacitor uses Aluminum oxide as the dielectric with relative permeability = 8. An effective surface area of 360 cm^2 gives a capacitance of $6 \mu\text{F}$. Calculate the field strength and the total dipole moment induced in oxide layer if a potential difference of 15 volts exist across the capacitor. **(Ans: (i) $3.535 \times 10^7 \text{ V/m}$ (ii) $33.4 \times 10 \text{ C - m}$)**
4. The radius of the Helium atom is about 0.55 AU. Calculate the polarisability of Helium and its relative permeability. The number of Helium atoms in a volume of 1 m is 2.7×10^{25} atoms. **(Ans: (i) $0.185 \times 10^{-40} \text{ Farad - m}^2$ (ii) 1.000057)**